IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

SEMICONDUCTOR DEVICES

Thursday, 5 December 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer question ONE. Answer any TWO of the remaining questions.

Calculators are permitted.

Formula sheet will be supplied with the exam paper.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible	First Marker(s) :	Z. Durrani
	Second Marker(s) :	K. Fobelets

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Constants

Permittivity of free space:	$\varepsilon_o = 8.85 \times 10^{-12}$ F/m
Permeability of free space:	$\mu_o = 4\pi \times 10^{-7}$ H/m
Intrinsic carrier concentration in Si:	$n_i = 1.45 \times 10^{10}$ cm^{-3} at $T = 300$ K
Dielectric constant of Si:	$\varepsilon_{Si} = 11$
Dielectric constant of SiO_2:	$\varepsilon_{ox} = 4$
Thermal voltage:	$kT/e = 0.026$ V at $T = 300$ K
Charge of an electron:	$e = 1.6 \times 10^{-19}$ C
Planck's constant:	$h = 6.63 \times 10^{-34}$ Js
Boltzmann constant:	$k = 1.38 \times 10^{-23}$ J/K
Band gap of Si:	$E_G = 1.12$ eV at $T = 300$ K
Electron affinity of Si:	$\chi = 4.05$ eV at $T = 300$ K
Effective density of states of Si:	$N_C = 3.2 \times 10^{19}$ cm^{-3} at $T = 300$ K $N_V = 1.8 \times 10^{19}$ cm^{-3} at $T = 300$ K

Formulae

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2}+V(x)\psi(x)=E\psi(x)$$ Schrödinger's equation in one dimension

$$f(E)=\frac{1}{1+\exp\left(\dfrac{E-E_f}{kT}\right)}$$ Fermi distribution

$$m_e^*=\frac{\hbar^2}{d^2E(k)/dk^2}$$ Effective mass

$$n=\frac{1}{\sqrt{2}\hbar^3}\left(\frac{m_e^*kT}{\pi}\right)^{3/2}e^{\left(-\frac{E_c-E_F}{kT}\right)}=N_ce^{\left(-\frac{E_c-E_F}{kT}\right)}$$ Concentration of electrons

$$p=\frac{1}{\sqrt{2}\hbar^3}\left(\frac{m_h^*kT}{\pi}\right)^{3/2}e^{\left(-\frac{E_F-E_v}{kT}\right)}=N_ve^{\left(-\frac{E_F-E_v}{kT}\right)}$$ Concentration of holes

$$\left.\begin{aligned}J_n(x)&=e\mu_n n(x)E(x)+eD_n\frac{dn(x)}{dx}\\J_p(x)&=e\mu_p p(x)E(x)-eD_p\frac{dp(x)}{dx}\end{aligned}\right\}$$ Drift and diffusion current densities in a semiconductor

$$I_{DS}=\frac{\mu C_{ox}W}{L}\left((V_{GS}-V_{th})V_{DS}-\frac{V_{DS}^2}{2}\right)$$ Current in a MOSFET

$$\left.\begin{aligned}J_n&=\frac{eD_n n_{p_0}}{L_n}\left(e^{\frac{eV}{kT}}-1\right)\\J_p&=\frac{eD_p p_{n_0}}{L_p}\left(e^{\frac{eV}{kT}}-1\right)\end{aligned}\right\}$$ Diffusion current densities in a pn-junction

$$V_0=\frac{kT}{e}\ln\left(\frac{N_AN_D}{n_i^2}\right)$$ Built-in voltage

$$c=c_0\exp\left(\frac{eV}{kT}\right)\text{ with }\begin{cases}c=p_n\text{ or }n_p\\c_0\text{ bulk minority carrier concentration}\end{cases}$$ Minority carrier injection under bias V

$$D=\frac{kT}{e}\mu$$ Einstein relation

$$w_n=\sqrt{\frac{2\varepsilon_0\varepsilon_r}{q}\frac{N_A}{N_AN_D+N_D^2}(V_0-V)}$$

$$w_p=\sqrt{\frac{2\varepsilon_0\varepsilon_r}{q}\frac{N_D}{N_AN_D+N_A^2}(V_0-V)}$$ Depletion width in the n and p-type region, under bias V

$$\delta c = \Delta c \exp\left(\frac{-x}{L}\right) \text{ with } \begin{cases} \delta c = \delta p_n \text{ or } \delta n_p \\ \Delta c \text{ the excess carrier concentration} \\ \text{at the edge of the depletion region} \end{cases}$$

Excess carrier concentration as a function of distance when recombination occurs – long layer approximation.

$$L = \sqrt{D\tau}$$

Diffusion length

$$C_{diff} = \frac{e}{kT} I\tau$$

Diffusion capacitance

$$i(t) = \frac{Q(t)}{\tau} + \frac{dQ(t)}{dt}$$

Time variation of current and charge

$$\frac{dE}{dx} = \frac{\rho(x)}{\varepsilon}$$

Poisson equation in 1 dimension

$$n = N_C \, exp\left(\frac{E_F - E_C}{kT}\right)$$

$$p = N_V \, exp\left(\frac{E_V - E_F}{kT}\right)$$

Carrier concentrations

1. [This question carries 40% marks]

a) By considering generation and recombination rates in a semiconductor, derive the law of mass action, $np = n_i^2$.

[4]

b) Four resistors are constructed using silicon and ideal Ohmic contacts. The resistors have the same geometry, shown in Fig. 1.1, however the doping density and type varies, shown in Table 1.1. Write down the room temperature electrical conductivity for the resistors σ_α, where α is the sample number, in ascending order.

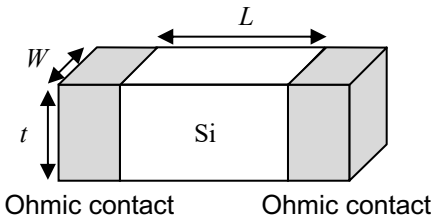


Figure 1.1. Resistor geometry

Sample number (α)	1	2	3	4
N_D (cm^{-3})	0	10^{17}	10^{14}	0
N_A (cm^{-3})	0	10^{16}	0	10^{14}

Table 1.1: Sample number and doping

[4]

c) Fig. 1.2 shows the variation of the collector current i_C in a BJT as a function of time, as the BJT is switched from cut-off to saturation.

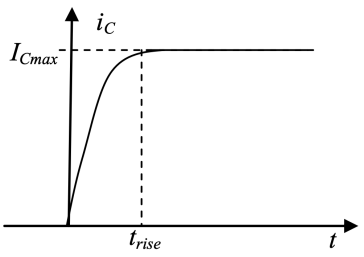


Figure 1.2

i) Explain why i_c saturates at a value I_{Cmax}.

ii) Explain the physical origin of the time delay t_{rise}.

[4 + 4]

d) In a semiconductor, a widely used approximation replaces the Fermi-Dirac distribution by the Boltzmann distribution. At temperature T = 300 K, for what range of energy E, measured with respect to the Fermi energy E_F, would such an approximation be valid, for <5% deviation from the Fermi-Dirac distribution?

[6]

e) State which type of current (drift, diffusion, generation, recombination) is most significant in a *pn* diode when it is biased under the following conditions:

i) Forward bias, near turn-on voltage.

ii) Forward bias, near zero bias.

iii) Reverse bias, well away from breakdown, within the depletion region.

[2+2+2]

f) Sketch the band diagram (E_c, E_v, E_F, E_G), from the emitter to the collector, for a p^+np bipolar junction transistor (BJT) with:

i) The BJT at zero bias.

[4]

ii) The BJT in cut-off.

[4]

iii) The BJT in active mode.

[4]

2. [This question carries 30% marks]

Consider a n^+p diode with the following parameters:

Donor doping density: $N_D = 2 \times 10^{19}$ cm^{-3}.
Acceptor doping density: $N_A = 10^{15}$ cm^{-3}.
Diffusion length of the minority carriers: $L_p = 10^{-2}$ cm (holes).
$L_n = 10^{-3}$ cm (electrons).
Diffusion constant of the minority carriers: $D_p = 10$ cm^2/s (holes).
$D_n = 20$ cm^2/s (electrons).
Area of the diode: $A = 10$ cm^2. The physical length of the n and p type region is exactly equal to the respective minority carrier diffusion lengths.

The temperature is: $T = 300$ K.

a) Draw the energy band diagram (including E_c, E_v, E_F and E_G) for this diode:

i) At equilibrium, zero applied bias.

[6]

ii) Under forward bias. Your diagram should also indicate the positive and negative terminals, and the direction of the net hole and net electron flux.

[6]

b) Determine the relative magnitude (greater than, less than, or equal) of the electron current density, J_n, vs. the hole current density, J_p, when the diode is forward biased.

[4]

c) Re-design the diode such that for the same bias the current density is increased by a factor of 10. You can only change one parameter, and you cannot change doping densities. Ensure your answer is realistic and state any approximations you make. Define and give the values of the revised parameters in your final answer.

[6]

Question continues on next page

d) Figure 2.1(a) shows the diode with external bias circuit. The diode is initially forward biased by voltage $v(t) = V$. At time $t = 0$, the polarity of this voltage is reversed to $-V$ (Fig. 2.1(b)). The minority carrier charge $Q_p(t)$ in the n region may be related to the minority carrier current for holes, $i_p(t)$ by the equation:

$$i_p(t) = \frac{Q_p(t)}{\tau_p} + \frac{Q_p(t)}{dt}$$

Derive an expression for $Q_p(t)$ as a function of time. [8]

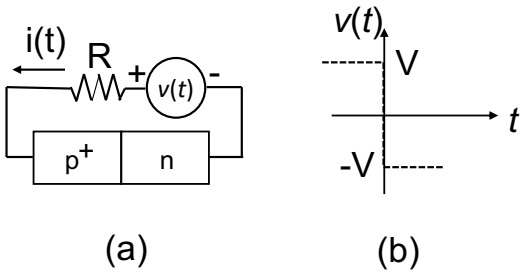


Figure 2.1

3. [This question carries 30% marks]

Figure 3.1 shows schematically a p^+np BJT where the depletion regions at the emitter-base (EB) and base-collector (BC) junctions along the x-axis are indicated by grey shaded areas (not to scale). The BJT has a constant cross-sectional area $A = 10\ \mu m^2$ and the width of the base region W_B is 2 μm The doping concentrations in the emitter, base and collector junctions are $N_{aE} = 10^{17}/cm^3$, $N_{dB} = 10^{16}/cm^3$ and $N_{aC} = 10^{16}/cm^3$ respectively. The BJT is to be biased in common-base mode, using supply voltages V_{EB} and V_{BC}.

a) Write the equation for charge neutrality across the EB depletion region, for $V_{EB} = 0$ V. Briefly justify any assumption you make.

[4]

b) Sketch the shape (you do not need to give values) of the charge concentration per unit volume $\rho(x)$ along the x-axis in the EB depletion region, for $V_{EB} = 0$ V.

[4]

c) By applying the Poisson equation in one-dimension (1D), sketch the shape (you do not need to give values) of the electric field $F(x)$ across the EB depletion region, and derive an expression for the maximum electric field F_{max} in the in the EB depletion region.

[6]

d) Hence, derive the following expression for the total EB depletion region width W:

$$W = \sqrt{\frac{2\varepsilon_{Si}\varepsilon_0}{e}\left(\frac{N_{aE}+N_{dB}}{N_{aE}N_{dB}}\right)(V_0 - V_{EB})}$$

Where V_0 is the built-in voltage in the EB junction.

[10]

e) The BJT is now biased in forward active mode, with $|V_{EB}| = |V_{BC}| = 0.5$ V. Calculate the emitter current I_E in the device. Your answer should take the depletion widths in the base region into account. You may neglect the current contribution from the emitter region. The diffusion constants for minority carrier holes and electrons are $D_p = 10\ cm^2/s$ and $D_n = 20\ cm^2/s$, respectively.

[6]

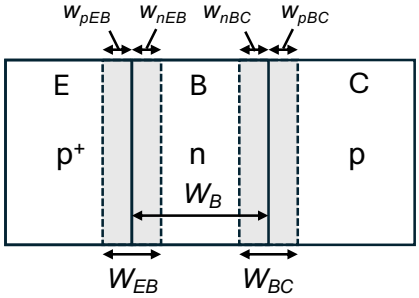


Figure 3.1

4. [This question carries 30% marks]

A cross-section through a *n*-channel, enhancement mode Si MOSFET is shown in Figure 4.1. The device has gate length L, gate width W, and gate oxide thickness t_{ox}. The drain-source and gate-source voltages are V_{DS} and V_{GS} respectively, the MOSFET threshold voltage is V_{th}, and the source is grounded. You may assume 'flat-band' conditions.

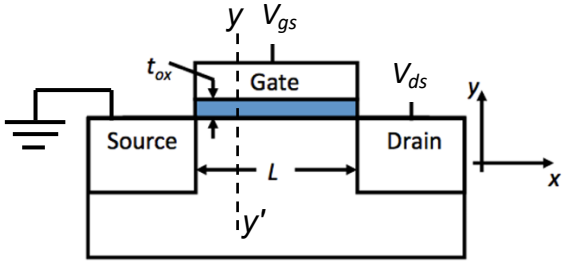


Figure 4.1

a) Sketch the energy band diagram (E_c, E_v, E_F, E_G) of the MOSFET, along the line *yy'* from the gate metal to the substrate, for the following conditions:

i) The MOSFET is in accumulation.

ii) The MOSFET is biased at threshold voltage $V_{GS} = V_{th}$.

iii) The MOSFET is biased at $V_{GS} > V_{th}$.

Your diagrams should indicate any applied biases.

[2+2+2]

b) Write a short note on the small signal equivalent circuit for a Si MOSFET. Your answer should discuss the location and significance of the internal capacitances, channel conductance, and transconductance of the MOSFET. State any impact of V_{ds} on the small-signal components.

[6]

Question continues on next page

c) Consider the 1-D hole minority carrier diffusion current density J_p(x), along the *x*-axis, in a semiconductor of fixed cross-sectional area *A*.

i) In the presence of an excess hole concentration $\delta p(x,t)$, derive the diffusion current continuity equation, in the presence of recombination, shown below:

$$\frac{\partial \delta p}{\partial t} = D_p \frac{\partial^2 \delta p}{\partial x^2} - \frac{\delta p}{\tau_p}$$

Where D_p is the diffusion constant and τ_p is the recombination time for holes.

[10]

ii) Hence, derive the steady state solution for $\delta p(x)$, for boundary conditions $\delta p(0) = \Delta p$ and $\delta p(X) = 0$, in the form:

$$\delta p(x) = \frac{\Delta p}{\sinh\left(\frac{-X}{L_p}\right)} \sinh\left(\frac{x - X}{L_p}\right)$$

Where L_p is the hole diffusion length.

[8]

[END]